InHand Networks

High Precision Tracking, Rugged for Harsh Environments

VT310 Vehicle Telematics Gateway

- LTE Cat-M/1/4
- GNSS
- IP66
- Dual CAN

1. Product Overview

The InHand VT310 is a rugged vehicle telematics gateway designed for reliable operation in challenging environments involving severe cold, high temperature, and/or water immersion.

Key Features:

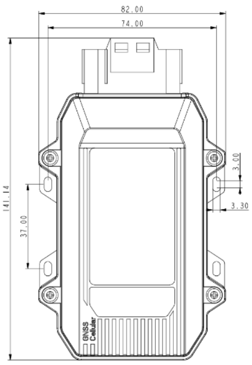
- **High precision GNSS:** GNSS + inertial navigation supports continuous accurate tracking
- **Driving behavior monitoring:** accelerometer and gyroscope enable monitoring of driving events and collisions
- **Vehicle-ready interfaces:** dual CAN bus, OBD-II/J1939/J1708 support, plus serial/I/O/Bluetooth/1-Wire
- **Continuous & reliable operation:** large cache (30K+ location records) and low power mode
- **Vehicle power adaptability:** wide voltage input 9–48V with IP66 protection

Core Technical Specifications

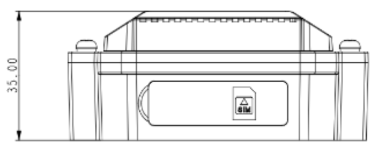
Technical Indicator	Specification
Cellular Network	LTE CAT-M / CAT1 / CAT4
Positioning Capability	GNSS multi-constellation positioning with inertial navigation (DR)
Vehicle Diagnostics	Supports OBD-II / J1939 / J1708
Cloud Platform Integration	Supports AWS IoT, Azure IoT, Aliyun IoT, Wialon, Traccar, and other platforms
Transport Protocols	TCP / UDP / HTTP / MQTT
Events and Alarms	Supports collision, motion, overspeed, IO change, ignition detection, and other alarms

Technical Indicator	Specification
Device Dimensions	Approx. 141 × 82 × 35 mm
Ingress Protection	IP66
Power Input Range	9-48 V DC
Interface Capability	Dual CAN, RS232, RS485, DI/DO, 1-Wire, and analog input
Operating Temperature	-40 to 85 C (external power); -20 to 60 C (battery)
Cache and Endurance	Supports 30K+ track record caching with built-in 1200 mAh battery

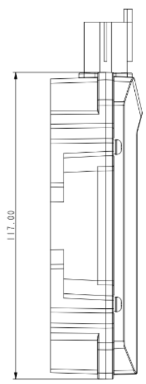
2. Product Dimensions



Front View



Interface View



Side View

Notes:

1. All dimensions are in millimeters (mm).
2. All dimensions are approximate and **for reference only**.
3. Drawings **must not be used for manufacturing**.
4. Dimensions are subject to part and manufacturing tolerances.
5. Specifications may change without prior notice.

3. Hardware Specifications

Category / Parameter	Specification
Network & Computing	
Network Type	LTE CAT-M / CAT1 / CAT4
CPU / Main Platform	Rugged multi-task system
Battery	1200 mAh
Protection Rating	IP66
Satellite Navigation & Inertial	
GNSS Receiver	GPS/GNSS/A-GNSS (multi-constellation)
Channel Count	31 channels
Initial Positioning Sensitivity	-162 dBm (initial positioning time 32 s)
Tracking Sensitivity	-156 dBm (hot start); -148 dBm (cold start)
Location Accuracy	2.5 m (CEP50)
Update Frequency	Default 1 Hz, max 10 Hz
Inertial Navigation	Supports continuous accurate tracking
Sensors	
Accelerometer Range	±2 / ±4 / ±8 / ±16 g
Gyroscope Range	±125 / ±250 / ±500 / ±1000 / ±2000 dps

Category / Parameter	Specification
Vehicle Interfaces	
CAN bus	2 channels
OBD & Diagnostics	J1708 / J1939 support
Serial Port	RS232 (1 channel)
RS485	RS485 (1 channel)
Ignition Signal	1 channel
Digital Input	4 channels
Digital Output	3 channels (max 300 mA)
1-Wire	1 channel (up to 4 sensors)
Analog Input	1 channel + 1 channel (per spec listing)
SIM Slot	2FF, push-in slot
LED Indicators	Cellular status, GNSS status
Power	
Working Voltage	9–48 V DC
Power Consumption	0.45 W / 0.55 W / 0.77 W (scenario-dependent)
Battery Rated Voltage	3.7 V
Battery Cut-off Voltage	4.2 V
Charging/Discharging Temp.	Charging: 0–45 ℃; Discharging: -20–60 ℃
Mechanical & Environment	

Category / Parameter	Specification
Shell Material	PC + ABS (engineering plastic + alloy)
Dimensions	Approximately 141 × 82 × 35 mm
Operating Temperature	-40 ~ 85 ℃ (connected to main power); -20 ~ 60 ℃ (battery)
Humidity	95% RH @ 50 ℃ (non-condensing)
ESD	IEC 61000-4-2 (4 kV test)
Certifications	
Compliance	CE, FCC, IC, PTCRB, E-Mark / E-Mark (per source listing)

4. Software Specifications

Category / Parameter	Specification
Cloud Platforms	
Supported Platforms	AWS IoT, Azure IoT, Aliyun IoT, Wialon, Traccar, GPSWox, WhiteLabel Tracking, ThingsBoard, Customer Platform
Network Features	
Protocols	TCP / UDP / HTTP / MQTT
Network Diagnostics	(as listed) connectivity / ping / traceroute / tcpdump / speed test (when supported)
Vehicle Data & Transparent Transmission	
Vehicle Data	OBD-II, J1939, J1708

Category / Parameter	Specification
Transparent Transmission	RS232/RS485 transparent + Modbus protocol data acquisition
Event Alarm & Reporting	
Event Alarm	Collision detection, motion detection, overspeed, IO change, ignition signal detection
Reporting	Support SMS or FlexAPI over TCP/UDP/MQTT
ELD	Supported
BLE Forwarding	Forward vehicle data via BLE
Configuration & Diagnostics	
Configuration Tool	RS232 or Bluetooth

5. Ordering Information

Model Rule

Model code: VT310-<WMNN>-B

<WMNN>: Cellular Type & Module

Product Models

Model	Cellular Type	Region

VT310-FQ58-B	LTE FDD: B1/B3/B5/B8; LTE TDD: B34/B38/B39/B40/B41; WCDMA: B1/B8; GSM: 900/1800 MHz	China / India
VT310-FQ53-B	LTE FDD: B1/B3/B5/B8/B20/B28; LTE CAT1	Europe / APAC
VT310-FQ08-B	LTE-FDD: B1/B2/B3/B4/B5/B7/B8/B12/B13/B18/ B19/B20/B25/B26/B28/66; LTE-TDD: B34/38/39/40/41; WCDMA: B1/2/4/5/6/8/19; GSM/EDGE: B2/3/5/8	Global

Cable Accessories

Cable	Order Code	Specifications
26 PIN All-in-one Test Cable	SCAB000 229	P1: 26 PIN female to connect to VT310; P2: open end requiring 9–48V adaptor. Suitable for engineering environments and indoor tests.
OBD-II 7 PIN All-in-one Cable	SCAB000 231	P1: 26 PIN female to VT310; P2: OBD-II male to vehicle; P3: ignition signal terminal. Suitable for heavy trucks with OBD-II interface and powers VT310 via interfaces.
OBD-II 26 PIN All-in-one Cable	SCAB000 232	P1: 26 PIN female to VT310; P2: OBD-II male to vehicle; includes I/O, RS232-1 and 1-Wire; P4: ignition signal terminal. Recommended for customers needing DI/DO/AI/1-Wire devices or vehicle-mounted controllers.

6. Contact Us

- **Website:** [InHand Networks]
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7. 26PIN Terminal Definition

Pin	Definition	Description
1	V-	Power negative terminal
2	GND	Ground
3	DI2	Digital input 2
4	DI4	Digital input 4
5	GND	Ground
6	DO2	Digital output 2
7	AI	Analog input
8	1-Wire	1-Wire interface
9	RS232_RX	RS232 receive
10	GND	Ground
11	CAN1_L	CAN1 low level
12	CAN2_L	CAN2 low level
13	J1708_B	J1708 B
14	V+	Power positive terminal
15	IGT	Ignition signal
16	DI1	Digital input 1

17	DI3	Digital input 3
18	GND	Ground
19	DO1	Digital output 1
20	DO3	Digital output 3
21	GND	Ground
22	RS232_TX	RS232 transmit
23	GND	Ground
24	CAN1_H	CAN1 high level
25	CAN2_H	CAN2 high level
26	J1708_A	J1708 A